

# A Logarithmic Barrier Function-Based Newton's Model for Solving Dynamic Quadratic Programming Subject to Inequality Constraints

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## ABSTRACT

This paper proposes a logarithmic barrier function-based Newton (LBFN) model for solving dynamic quadratic programming problems subject to inequality constraints (DQP-IC) in real time. Unlike conventional Karush–Kuhn–Tucker (KKT)-based schemes that require auxiliary dual variables, the LBFN model embeds a parametric, adaptive logarithmic barrier directly into the objective, so that constraint enforcement and optimization proceed within a single continuous-time dynamical system. A prediction-correction mechanism anticipates the temporal drift of the time-varying optimal solution and couples it with a Newton-type correction term, yielding global asymptotic stability from arbitrary feasible or infeasible initial conditions and an exponentially decaying tracking error with a user-tunable convergence rate. We further establish the well-posedness of the resulting dynamics, derive an explicit bound on the distance between the tracked trajectory and the true time-varying optimum, and show that the per-step computational cost scales comparably to a single Newton step of a static quadratic program. Numerical simulations against active-set, sequential quadratic programming, SeDuMi-based interior-point, continuous gradient-descent, and Newton-flow baselines demonstrate consistently lower tracking error and superior constraint satisfaction under fast parameter drift. A robot navigation case study, tracking a moving target while avoiding seven dynamic obstacles in real time, further illustrates the practical relevance of the proposed approach.

## 1 Introduction

Operating in highly dynamic environments, modern engineering and control systems demand instantaneous adaptation of optimal decisions to rapidly changing conditions<sup>1–3</sup>. This necessity for real-time adaptive programming is particularly critical in domains such as robotic autonomous navigation<sup>4–6</sup> and intelligent traffic management systems<sup>7,8</sup>. Effective control of these systems hinges on the continuous resolution of programming problems with time-varying objectives and constraints. As applications increase in scale and complexity, the development of practical methodologies capable of accurately tracking optimal solutions under dynamic variations has become indispensable across scientific and engineering domains, from aerospace target tracking and microgrid control<sup>9,10</sup> to large-scale nonconvex optimization on networked systems<sup>11</sup>.

Conventional strategies primarily address static programming problem formulations, where parameters remain fixed throughout the solution process<sup>12,13</sup>. Techniques like gradient-flow dynamics utilize first-order descent principles for unconstrained minimization<sup>14,15</sup>, while Newton-type flows accelerate convergence by incorporating second-order curvature information<sup>16,17</sup>. Constrained problems often employ active set strategies<sup>18,19</sup> and sequential quadratic programming<sup>20,21</sup>. However, the above-mentioned approaches are proven effective in dealing with static programming problems; these approaches encounter limitations when deployed in dynamic programming problems with persistent parameter drifts, because they lack an effective predictive mechanism to compensate for the solution drifts<sup>22,23</sup>.

A critical challenge arises because static solvers lack temporal awareness. When applied to dynamic systems, treating each programming instance as an independent snapshot, they inevitably generate substantial tracking errors<sup>24–26</sup>. For example, consider a robotic manipulator that moves between points while avoiding dynamically moving obstacles<sup>27,28</sup>. If its path planner uses a static optimizer without prediction, it would solve the problem only at the current instant based on the obstacle's current position. However, during the finite computation time, e.g., 50 ms, required to compute the optimal solution, both the robot and the obstacles have moved. The solution computed at  $t_k$  is therefore optimized for a past state ( $t_k$ ) and applied at a future state ( $t_k$  + computation time), resulting in sub-optimality or, worse, constraint violation if the obstacle moved unexpectedly close during that interval<sup>29,30</sup>. Similarly, in automated lane-changing scenarios for autonomous vehicles, a static planner reacting only to the instantaneous relative positions of surrounding vehicles without anticipating their motion trends could fail to generate a safe and efficient trajectory when neighboring vehicles accelerate or decelerate unexpectedly during the planning computation<sup>31</sup>.

These limitations highlight two critical gaps: a lack of inherent mechanism to compensate for the temporal drift of the optimal solution between the time computation starts and ends, and a reliance on strategies for handling inequality constraints with KKT-based approaches that are computationally expensive to recompute frequently or that introduce additional variables, complicating the real-time solution process<sup>32,33</sup>.

Beyond robotics and transportation, this same drift-and-constraint challenge underlies a broad class of continuously reconfigured decision problems in modern computing systems, including receding-horizon and model-predictive controllers that must re-solve quadratic subproblems every sampling interval, and edge- or embedded-computing pipelines that allocate limited resources under time-varying, hard operational bounds. In all such settings, an optimization primitive that both tolerates aggressive parameter drift and avoids the bookkeeping overhead of auxiliary dual variables is of direct practical value, since it lowers the per-iteration computational burden that ultimately governs achievable sampling rates.

To address these limitations, this paper proposes a logarithmic barrier function-based Newton's (LBFN) Model for dynamic inequality-constrained quadratic programming. Departing from the conventional paradigm of converting inequalities into equality systems via KKT conditions, which requires the introduction of dual variables, the LBFN model employs parametric adaptive log-barrier functions to directly incorporate dynamic inequality constraints. This formulation fundamentally avoids the need for auxiliary variables. The proposed model integrates a prediction mechanism that estimates the future drift of the optimal solution using time derivatives of the problem parameters, along with a Newton-based correction dynamics that drives the solution toward the optimum of the barrier-augmented objective while ensuring continuous constraint satisfaction throughout the evolution. The main contributions of this work are summarized as follows:

- A novel LBFN model is constructed for real-time solving of dynamic quadratic programming subject to inequality constraints (DQP-IC), which employs parametric adaptive log-barrier functions instead of KKT-based constraint handling.
- The LBFN model exhibits initialization insensitivity, enabling convergence to the optimal trajectory from arbitrary initial points (feasible or infeasible), while maintaining exponentially decaying tracking errors with a tunable convergence rate. We formally establish the well-posedness of the model and derive its per-iteration computational complexity.
- Theoretical analysis, numerical simulations, and a robotic navigation application are presented to validate the effectiveness and advantages of the proposed approach, together with a discussion of the model's limitations and directions for future work.

**Notation:** For any dynamic vector  $\mathbf{v}(t) \in \mathbb{R}^n$ , the Euclidean norm is expressed as  $\|\mathbf{v}(t)\|_2$ . Consider a function  $\tilde{h}(\mathbf{v}(t), t)$ . The optimal value of this function is denoted by  $\tilde{h}(\mathbf{v}^*(t), t)$ , where  $\mathbf{v}^*(t)$  signifies the optimal trajectory of the decision variable  $\mathbf{v}(t)$ . The gradient of  $\tilde{h}$  with respect to  $\mathbf{v}(t)$  is represented by  $\nabla_{\mathbf{v}}\tilde{h}(\mathbf{v}(t), t)$ , and its partial derivative with respect to time  $t$  is given by  $\partial_t\tilde{h}(\mathbf{v}(t), t)$ . The Hessian matrix of  $\tilde{h}$  with respect to  $\mathbf{v}(t)$  is denoted by  $\nabla_{\mathbf{v}\mathbf{v}}\tilde{h}(\mathbf{v}(t), t)$ . Furthermore, the partial derivative of the gradient  $\nabla_{\mathbf{v}}\tilde{h}(\mathbf{v}(t), t)$  with respect to  $t$  is written as  $\nabla_{\mathbf{v}t}\tilde{h}(\mathbf{v}(t), t)$ . The  $i$ -th component of the vector  $\mathbf{v}(t)$  is indicated by  $\mathbf{v}_i(t)$ , and  $\mathbb{S}_{++}^n$  denotes the cone of  $n \times n$  symmetric positive-definite matrices.

## 2 Preliminary and Scheme Formulation

In this part, a dynamic quadratic programming subject to inequality constraints and the LBFN model preliminaries are provided in detail.

### 2.1 Construction of DQP-IC Problem

Consider a DQP-IC problem formulated as:

$$\begin{aligned} \underset{\mathbf{v}(t) \in \mathbb{R}^n}{\text{minimize}} \quad & \tilde{h}(\mathbf{v}(t), t) = \frac{1}{2} \mathbf{v}^\top(t) \mathcal{M}(t) \mathbf{v}(t) + \mathbf{g}^\top(t) \mathbf{v}(t) \\ \text{subject to} \quad & \mathcal{N}(t) \mathbf{v}(t) \leq \mathbf{d}(t) \end{aligned} \quad (1)$$

where  $\mathbf{v}(t) \in \mathbb{R}^n$  is a decision variable.  $t \geq 0$  denotes continuous time.  $\mathcal{M}(t) \in \mathbb{S}_{++}^n$  is a symmetric positive-definite matrix with  $\mathcal{M}(t) \succeq m_f \mathcal{I}_n$  for some  $m_f > 0$  (uniform strong convexity).  $\mathbf{g}(t) \in \mathbb{R}^n$  is a dynamic linear coefficient vector.  $\mathcal{N}(t) \in \mathbb{R}^{p \times n}$  and  $\mathbf{d}(t) \in \mathbb{R}^p$  define  $p$  linear inequality constraints, with  $\mathcal{N}_i(t)$  denoting the  $i$ -th row of  $\mathcal{N}(t)$  and  $d_i(t)$  the  $i$ -th entry of  $\mathbf{d}(t)$ . All problem data  $\mathcal{M}(t)$ ,  $\mathbf{g}(t)$ ,  $\mathcal{N}(t)$ ,  $\mathbf{d}(t)$  and their first time derivatives are assumed continuous and available in closed form. The feasible set  $\mathcal{F}(t) = \{\mathbf{v}(t) \in \mathbb{R}^n : \mathcal{N}(t) \mathbf{v}(t) \leq \mathbf{d}(t)\}$  satisfies Slater's condition for all  $t \geq 0$ .

To handle constraints, we introduce a dynamic barrier parameter  $w(t) > 0$  with  $w(t) = w(0)e^{\gamma_w t}$ ,  $\gamma_w > 0$ , so that  $\lim_{t \rightarrow \infty} w(t) = \infty$ , and a dynamic slack variable  $r(t) > 0$  with  $r(t) = r(0)e^{-\gamma_r t}$ ,  $\gamma_r > 0$ , ensuring  $r(t) \rightarrow 0$  asymptotically. The modified objective

$\Upsilon(\mathbf{v}(t), w(t), r(t), t)$  is defined as:

$$\Upsilon(\mathbf{v}(t), w(t), r(t), t) = \hbar(\mathbf{v}(t), t) - \frac{1}{w(t)} \sum_{i=1}^p \log \left( r(t) - (\mathcal{A}_i(t)\mathbf{v}(t) - d_i(t)) \right) \quad (2)$$

The domain  $\widehat{\mathcal{D}}(t) = \{\mathbf{v}(t) \in \mathbb{R}^n : \mathcal{A}_i(t)\mathbf{v}(t) - d_i(t) < r(t), i = 1, \dots, p\}$  is non-empty when  $r(0) > \max_i(\mathcal{A}_i(0)\mathbf{v}(0) - d_i(0))$ .

## 2.2 General Framework of LBFN Model

Then, an implicit evolution dynamics is constructed as:

$$\dot{\mathbf{v}}(t) = -\nabla_{\mathbf{v}\mathbf{v}}^{-1} \Upsilon [\alpha \nabla_{\mathbf{v}} \Upsilon + \nabla_{\mathbf{v}r} \dot{r} + \nabla_{\mathbf{v}w} \dot{w} + \nabla_{\mathbf{v}t} \Upsilon] \quad (3)$$

where  $\alpha > 0$  is a convergence rate parameter.  $\nabla_{\mathbf{v}\mathbf{v}} \Upsilon$  is the Hessian of  $\Upsilon$  w.r.t.  $\mathbf{v}(t)$ .  $\nabla_{\mathbf{v}} \Upsilon$ ,  $\nabla_{\mathbf{v}r} \Upsilon$ ,  $\nabla_{\mathbf{v}w} \Upsilon$ , and  $\nabla_{\mathbf{v}t} \Upsilon$  are partial derivatives.

Define  $\delta_i(\mathbf{v}(t), t) = r(t) - (\mathcal{A}_i(t)\mathbf{v}(t) - d_i(t))$ , and the following definitions are given:

$$\nabla_{\mathbf{v}} \Upsilon = \mathcal{M}(t)\mathbf{v}(t) + \mathbf{g}(t) + \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i(\mathbf{v}(t), t)} \quad (4)$$

$$\nabla_{\mathbf{v}\mathbf{v}} \Upsilon = \mathcal{M}(t) + \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top \mathcal{A}_i(t)}{\delta_i^2(\mathbf{v}(t), t)} \quad (5)$$

$$\nabla_{\mathbf{v}r} \Upsilon = \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i^2(\mathbf{v}(t), t)} \quad (6)$$

$$\nabla_{\mathbf{v}w} \Upsilon = -\frac{1}{w^2(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i(\mathbf{v}(t), t)} \quad (7)$$

$$\nabla_{\mathbf{v}t} \Upsilon = \dot{\mathcal{M}}(t)\mathbf{v}(t) + \dot{\mathbf{g}}(t) + \frac{1}{w(t)} \sum_{i=1}^p \frac{\dot{\mathcal{A}}_i(t)^\top}{\delta_i(\mathbf{v}(t), t)} + \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top (\dot{\mathcal{A}}_i(t)\mathbf{v}(t) - \dot{d}_i(t))}{\delta_i^2(\mathbf{v}(t), t)} \quad (8)$$

Substituting into (3), it is eventually obtained the following LBFN model:

$$\begin{aligned} \dot{\mathbf{v}}(t) = & - \left( \mathcal{M}(t) + \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top \mathcal{A}_i(t)}{\delta_i^2(\mathbf{v}(t), t)} \right)^{-1} \left( \alpha \left( \mathcal{M}(t)\mathbf{v}(t) + \mathbf{g}(t) + \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i(\mathbf{v}(t), t)} \right) \right. \\ & \left. + \left( \frac{1}{w(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i^2(\mathbf{v}(t), t)} \right) \dot{r}(t) + \left( -\frac{1}{w^2(t)} \sum_{i=1}^p \frac{\mathcal{A}_i(t)^\top}{\delta_i(\mathbf{v}(t), t)} \right) \dot{w}(t) + \nabla_{\mathbf{v}t} \Upsilon \right) \end{aligned} \quad (9)$$

where  $\dot{r}(t) = -\gamma_r r(t)$  and  $\dot{w}(t) = \gamma_w w(t)$  follow directly by differentiating  $r(t) = r(0)e^{-\gamma_r t}$  and  $w(t) = w(0)e^{\gamma_w t}$ , respectively.

While (1) restricts  $\hbar$  to be quadratic so that (9) admits the closed form above and the theoretical analysis of Section 3 applies directly, the underlying implicit dynamics (3) are not intrinsically limited to quadratic objectives: for any  $\hbar(\mathbf{v}(t), t)$  that is twice continuously differentiable and  $m_f$ -strongly convex in  $\mathbf{v}(t)$  for every  $t$ , substituting the general  $\nabla_{\mathbf{v}} \hbar$ ,  $\nabla_{\mathbf{v}\mathbf{v}} \hbar$ , and  $\nabla_{\mathbf{v}t} \hbar$  for  $\mathcal{M}(t)\mathbf{v}(t) + \mathbf{g}(t)$ ,  $\mathcal{M}(t)$ , and  $\mathcal{M}(t)\mathbf{v}(t) + \dot{\mathbf{g}}(t)$  in (4)–(8) yields the same prediction-correction structure, and Lemma 1, Theorem 1, and Corollary 1 continue to hold verbatim with  $\hbar$  in place of the quadratic form, provided  $\hbar(\cdot, t)$  remains  $m_f$ -strongly convex for every  $t$ . Section 4.2 uses this more general instantiation for a non-quadratic potential-field objective that is only locally, not globally, strongly convex (Section 5); that case study therefore demonstrates the LBFN mechanism empirically beyond the setting formally covered by Theorem 1, rather than constituting an additional instance to which the guarantees provably apply.

### 3 Theoretical Analysis

This part provides the theoretical analysis for the LBFN model (9), including its well-posedness, global convergence, and computational complexity.

**Lemma 1** (Well-posedness). *For every  $t \geq 0$  and every  $\mathbf{v}(t) \in \widehat{\mathcal{D}}(t)$ , the Hessian  $\nabla_{\mathbf{v}\mathbf{v}}\Upsilon(\mathbf{v}(t), \mathbf{w}(t), \mathbf{r}(t), t)$  in (5) is symmetric positive definite, and the right-hand side of (9) is locally Lipschitz continuous in  $\mathbf{v}(t)$  on  $\widehat{\mathcal{D}}(t)$ . Consequently, for any  $\mathbf{v}(0) \in \widehat{\mathcal{D}}(0)$  the LBFN model (9) admits a unique solution trajectory  $\mathbf{v}(t)$  that remains in  $\widehat{\mathcal{D}}(t)$  for all  $t \geq 0$ .*

*Proof.* By assumption  $\mathcal{M}(t) \succeq m_f \mathcal{I}_n \succ 0$ . For  $\mathbf{v}(t) \in \widehat{\mathcal{D}}(t)$ ,  $\delta_i(\mathbf{v}(t), t) > 0$  for every  $i$ , so each rank-one term  $\mathcal{A}_i(t)^\top \mathcal{A}_i(t) / \delta_i^2(\mathbf{v}(t), t)$  in (5) is symmetric positive semidefinite; a sum of a positive-definite matrix and positive-semidefinite matrices is positive definite, so  $\nabla_{\mathbf{v}\mathbf{v}}\Upsilon \succeq m_f \mathcal{I}_n \succ 0$  and is therefore invertible with  $\|\nabla_{\mathbf{v}\mathbf{v}}^{-1}\Upsilon\|_2 \leq 1/m_f$ . Because  $\Upsilon$  is twice continuously differentiable on  $\widehat{\mathcal{D}}(t)$  (all barrier terms are smooth away from the boundary  $\delta_i = 0$ ), the right-hand side of (9), being a composition of continuously differentiable maps with the uniformly bounded inverse  $\nabla_{\mathbf{v}\mathbf{v}}^{-1}\Upsilon$ , is locally Lipschitz in  $\mathbf{v}(t)$  on any compact subset of  $\widehat{\mathcal{D}}(t)$ . Existence and uniqueness of a local solution then follow from the Picard–Lindelöf theorem on any compact subset of  $\widehat{\mathcal{D}}(t)$ ; Corollary 1 below shows this local solution never approaches the boundary  $\delta_i \rightarrow 0^+$  in finite time and hence extends to all  $t \geq 0$ .  $\square$

**Theorem 1** (Global exponential convergence). *Consider the LBFN model (9) for the DQP-IC problem (1). Define the Lyapunov function candidate:*

$$V(t) = \frac{1}{2} \|\nabla_{\mathbf{v}}\Upsilon(\mathbf{v}(t), \mathbf{w}(t), \mathbf{r}(t), t)\|_2^2 \quad (10)$$

*For any initial condition  $\mathbf{v}(0) \in \widehat{\mathcal{D}}(0)$ ,  $V(t)$  satisfies  $\dot{V}(t) = -2\alpha V(t)$  for as long as the solution exists, and consequently*

$$\lim_{t \rightarrow \infty} \|\nabla_{\mathbf{v}}\Upsilon(\mathbf{v}(t), \mathbf{w}(t), \mathbf{r}(t), t)\|_2 = 0, \quad (11)$$

*i.e., the tracking error vanishes uniformly for every initial condition, regardless of how far  $\mathbf{v}(0)$  starts from optimality.*

*Proof.* By Lemma 1,  $\nabla_{\mathbf{v}\mathbf{v}}\Upsilon$  is invertible along the trajectory, so (3) is well defined. The time derivative of  $V(t)$  is computed as:

$$\dot{V}(t) = \nabla_{\mathbf{v}}\Upsilon^\top \left( \frac{d}{dt} \nabla_{\mathbf{v}}\Upsilon \right) \quad (12)$$

The total derivative  $\frac{d}{dt} \nabla_{\mathbf{v}}\Upsilon$  expands to:

$$\frac{d}{dt} \nabla_{\mathbf{v}}\Upsilon = \nabla_{\mathbf{v}\mathbf{v}}\Upsilon \dot{\mathbf{v}} + \nabla_{\mathbf{v}\mathbf{r}}\Upsilon \dot{\mathbf{r}} + \nabla_{\mathbf{v}\mathbf{w}}\Upsilon \dot{\mathbf{w}} + \nabla_{\mathbf{v}t}\Upsilon \quad (13)$$

Substitute the dynamics (3) into (13):

$$\frac{d}{dt} \nabla_{\mathbf{v}}\Upsilon = \nabla_{\mathbf{v}\mathbf{v}}\Upsilon \left( -\nabla_{\mathbf{v}\mathbf{v}}^{-1}\Upsilon (\alpha \nabla_{\mathbf{v}}\Upsilon + \nabla_{\mathbf{v}\mathbf{r}}\Upsilon \dot{\mathbf{r}} + \nabla_{\mathbf{v}\mathbf{w}}\Upsilon \dot{\mathbf{w}} + \nabla_{\mathbf{v}t}\Upsilon) \right) + \nabla_{\mathbf{v}\mathbf{r}}\Upsilon \dot{\mathbf{r}} + \nabla_{\mathbf{v}\mathbf{w}}\Upsilon \dot{\mathbf{w}} + \nabla_{\mathbf{v}t}\Upsilon \quad (14)$$

Simplify the expression:

$$\frac{d}{dt} \nabla_{\mathbf{v}}\Upsilon = -\alpha \nabla_{\mathbf{v}}\Upsilon - \nabla_{\mathbf{v}t}\Upsilon + \nabla_{\mathbf{v}t}\Upsilon = -\alpha \nabla_{\mathbf{v}}\Upsilon \quad (15)$$

The cancellation occurs because the prediction terms  $(\nabla_{\mathbf{v}\mathbf{r}}\Upsilon \dot{\mathbf{r}} + \nabla_{\mathbf{v}\mathbf{w}}\Upsilon \dot{\mathbf{w}} + \nabla_{\mathbf{v}t}\Upsilon)$  are exactly compensated by the additional derivatives in the total derivative. This is a key feature of the prediction-correction structure.

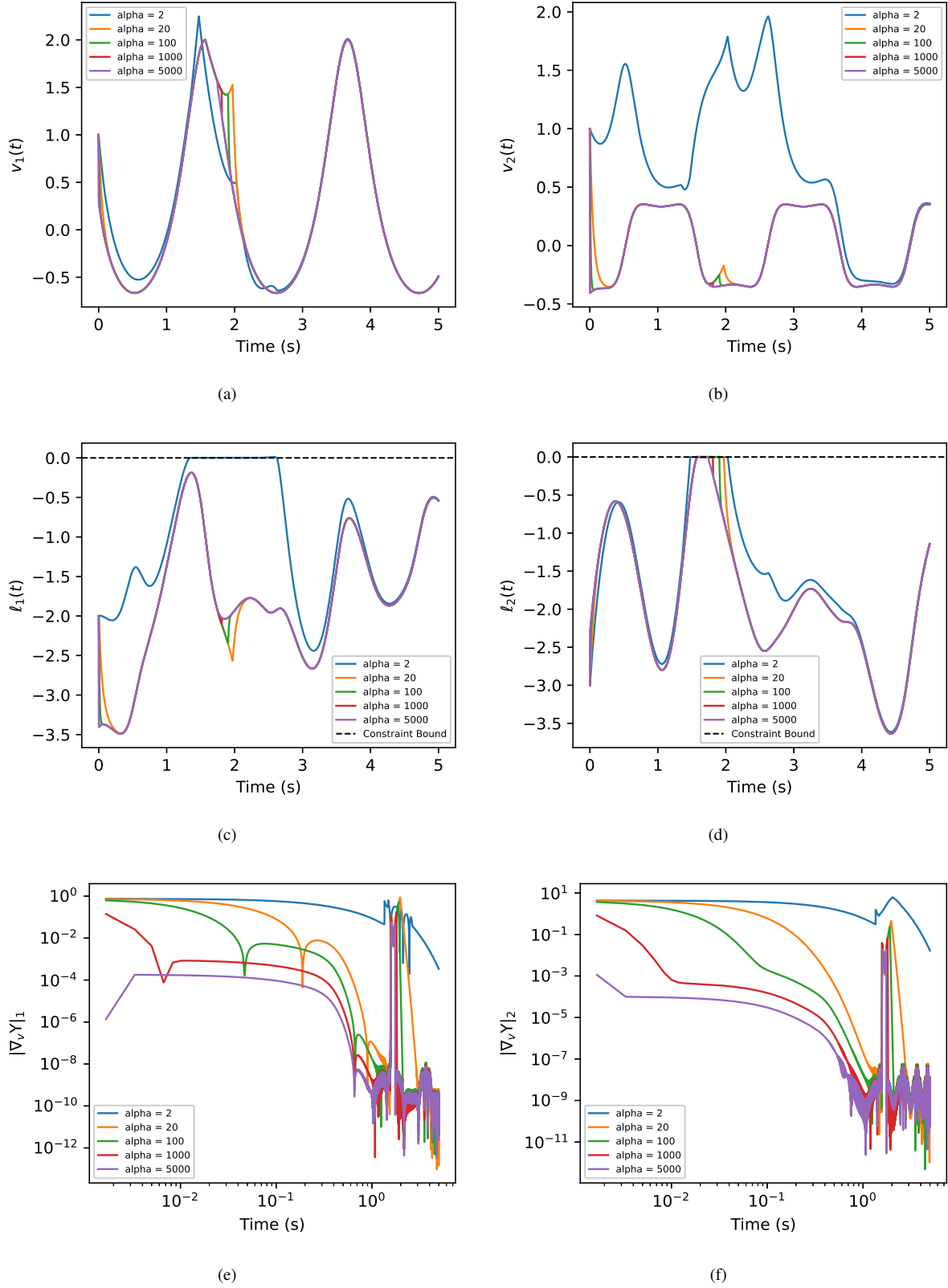
Substitute (15) into (12):

$$\dot{V}(t) = \nabla_{\mathbf{v}}\Upsilon^\top (-\alpha \nabla_{\mathbf{v}}\Upsilon) = -2\alpha V(t) \quad (16)$$

Solving this ODE gives<sup>34</sup>:

$$V(t) = V(0)e^{-2\alpha t} \quad (17)$$

which implies  $\lim_{t \rightarrow \infty} V(t) = 0$ , proving (11).  $\square$



**Figure 1.** Simulative outcomes of the presented LBFN model (9) to solve the optimal solution trajectory of the DQP-IC problem (22), with convergent parameter  $\alpha$  varies from 2 to 5000,  $\gamma_w = 10$  and  $\gamma_r = 5$ . (a) Convergence behaviors of  $v_1(t)$ . (b) Convergence behaviors of  $v_2(t)$ . (c) Constraint behaviors of  $g_1(t) = \sin(2t)v_1(t) + v_2(t) - (2 + \cos(2t))$ . (d) Constraint behaviors of  $g_2(t) = -\cos(2t)v_1(t) + (2t/10)v_2(t) - (2 - \sin(4t))$ . (e) The first entry of the absolute gradient error. (f) The second entry of the absolute gradient error.

**Corollary 1** (Tracking error relative to the true optimal trajectory). *Let  $\mathbf{v}_\Upsilon^*(t) := \arg \min_{\mathbf{v}} \Upsilon(\mathbf{v}, w(t), r(t), t)$  denote the instantaneous minimizer of the barrier-augmented objective and  $\mathbf{v}^*(t)$  the true minimizer of (1). Then*

$$\|\mathbf{v}(t) - \mathbf{v}^*(t)\|_2 \leq \frac{\sqrt{2V(0)}}{m_f} e^{-\alpha t} + \sqrt{\frac{2p}{m_f w(0)}} e^{-\gamma_w t/2}, \quad (18)$$

so  $\mathbf{v}(t)$  converges to the true time-varying optimal trajectory at rate  $\min(\alpha, \gamma_w/2)$ , and in particular  $\mathbf{v}(t)$  remains within a bounded, shrinking distance of  $\mathbf{v}^*(t)$  for all  $t \geq 0$  (hence never approaches  $\partial \widehat{\mathcal{D}}(t)$  in finite time, completing the proof of Lemma 1).

*Proof.* By Lemma 1,  $\Upsilon(\cdot, w(t), r(t), t)$  is  $m_f$ -strongly convex in  $\mathbf{v}$  for every  $t$ , so the standard strong-convexity gradient inequality gives  $\|\mathbf{v}(t) - \mathbf{v}_\Upsilon^*(t)\|_2 \leq \frac{1}{m_f} \|\nabla_{\mathbf{v}} \Upsilon(\mathbf{v}(t), t)\|_2 = \frac{\sqrt{2V(t)}}{m_f} = \frac{\sqrt{2V(0)}}{m_f} e^{-\alpha t}$  by Theorem 1. Separately,  $\mathbf{v}_\Upsilon^*(t)$  is feasible for (1), and the classical log-barrier suboptimality bound<sup>34</sup> gives  $\tilde{h}(\mathbf{v}_\Upsilon^*(t), t) - \tilde{h}(\mathbf{v}^*(t), t) \leq p/w(t)$ ; since  $\tilde{h}(\cdot, t)$  is itself  $m_f$ -strongly convex (as  $\mathcal{M}(t) \succeq m_f \mathcal{J}_n$ ) and  $\mathbf{v}^*(t)$  minimizes  $\tilde{h}(\cdot, t)$  over the feasible set,  $\frac{m_f}{2} \|\mathbf{v}_\Upsilon^*(t) - \mathbf{v}^*(t)\|_2^2 \leq \tilde{h}(\mathbf{v}_\Upsilon^*(t), t) - \tilde{h}(\mathbf{v}^*(t), t) \leq p/w(t)$ , i.e.  $\|\mathbf{v}_\Upsilon^*(t) - \mathbf{v}^*(t)\|_2 \leq \sqrt{2p/(m_f w(t))} = \sqrt{2p/(m_f w(0))} e^{-\gamma_w t/2}$ . Combining both bounds via the triangle inequality  $\|\mathbf{v}(t) - \mathbf{v}^*(t)\|_2 \leq \|\mathbf{v}(t) - \mathbf{v}_\Upsilon^*(t)\|_2 + \|\mathbf{v}_\Upsilon^*(t) - \mathbf{v}^*(t)\|_2$  yields (18). Since the right-hand side of (18) is finite and continuous for all  $t \geq 0$ ,  $\mathbf{v}(t)$  remains a bounded distance from the strictly feasible  $\mathbf{v}_\Upsilon^*(t)$  (in particular, bounded away from  $\partial \widehat{\mathcal{D}}(t)$  on every compact time interval), so the local solution of Lemma 1 cannot cease to exist at any finite  $t$ .  $\square$

### 3.1 Computational Complexity Analysis

We next characterize the per-instant computational cost of evaluating (9), which governs its suitability for real-time implementation. Let  $n$  denote the decision-variable dimension and  $p$  the number of inequality constraints. Forming the Hessian  $\nabla_{\mathbf{v}\mathbf{v}} \Upsilon$  in (5) requires summing  $p$  rank-one outer products  $\mathcal{A}_i(t)^\top \mathcal{A}_i(t)$ , each costing  $O(n^2)$  operations, for a total of  $O(pn^2)$ ; forming the gradient terms (4)–(8) costs  $O(pn)$ . Since  $\nabla_{\mathbf{v}\mathbf{v}} \Upsilon$  is symmetric positive definite by Lemma 1, the linear solve implicit in (3) can be carried out by Cholesky factorization at a cost of  $O(n^3)$  rather than by an explicit  $O(n^3)$  matrix inversion followed by a matrix–vector product, and the factorization can be reused across the right-hand-side terms sharing the same Hessian at a given instant<sup>34</sup>. The overall per-instant complexity of the LBFN update is therefore

$$O(n^3 + pn^2), \quad (19)$$

which is dominated by the  $O(n^3)$  factorization step whenever  $p = O(n)$ , matching the cost of a single Newton step for a static quadratic program of the same size. This compares favorably with KKT-based dynamic solvers, whose per-instant cost includes forming and factorizing an augmented system of dimension  $n + p$  (or  $n + 2p$  when slack variables are introduced), and with active-set methods, whose worst-case combinatorial search over constraint activity has cost that grows with the number of candidate active sets rather than polynomially in  $n$  and  $p$ . When  $n$  is large, the  $O(n^3)$  factorization cost can be further reduced using rank-one Hessian updates via the Sherman–Morrison–Woodbury identity or warm-started factorizations that exploit the continuity of  $\nabla_{\mathbf{v}\mathbf{v}} \Upsilon(t)$  across nearby time instants; we return to this point in Section 5.

## 4 Experimental Results

This section provides two examples to demonstrate the effectiveness and advantages of the LBFN model.

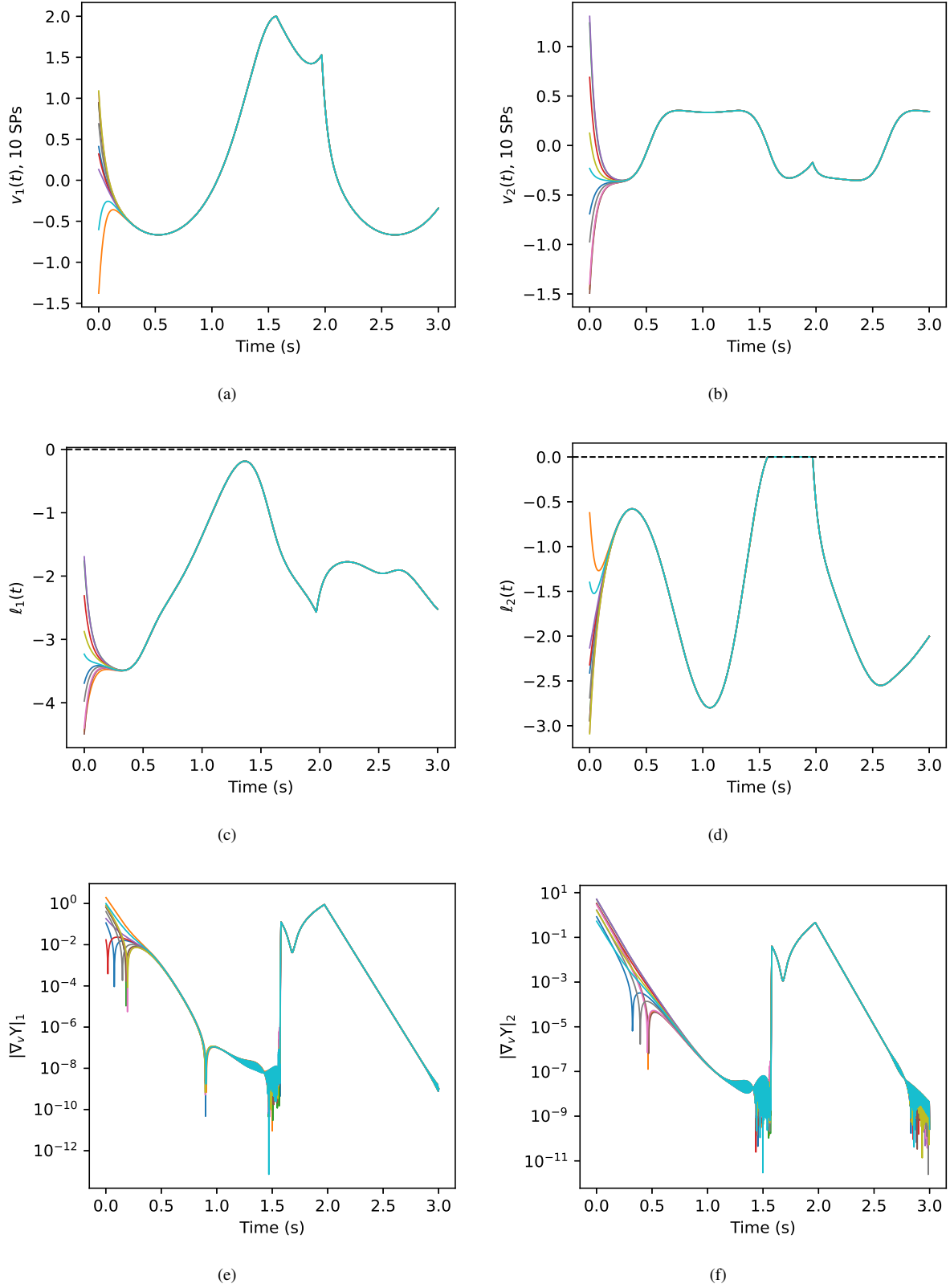
### 4.1 Numerical Example

The quadratic objective function is designed to be strictly convex and dynamic through its parameters:

$$\tilde{h}(\mathbf{v}(t), t) = \frac{1}{2} \begin{bmatrix} v_1(t) & v_2(t) \end{bmatrix} \underbrace{\begin{bmatrix} 1 + 0.5 \sin(3t) & 0 \\ 0 & 2 + \cos(6t) \end{bmatrix}}_{\mathcal{P}(t)} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} + \underbrace{\begin{bmatrix} \sin(3t) & \cos(3t) \end{bmatrix}}_{\mathbf{n}^T(t)} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} \quad (20)$$

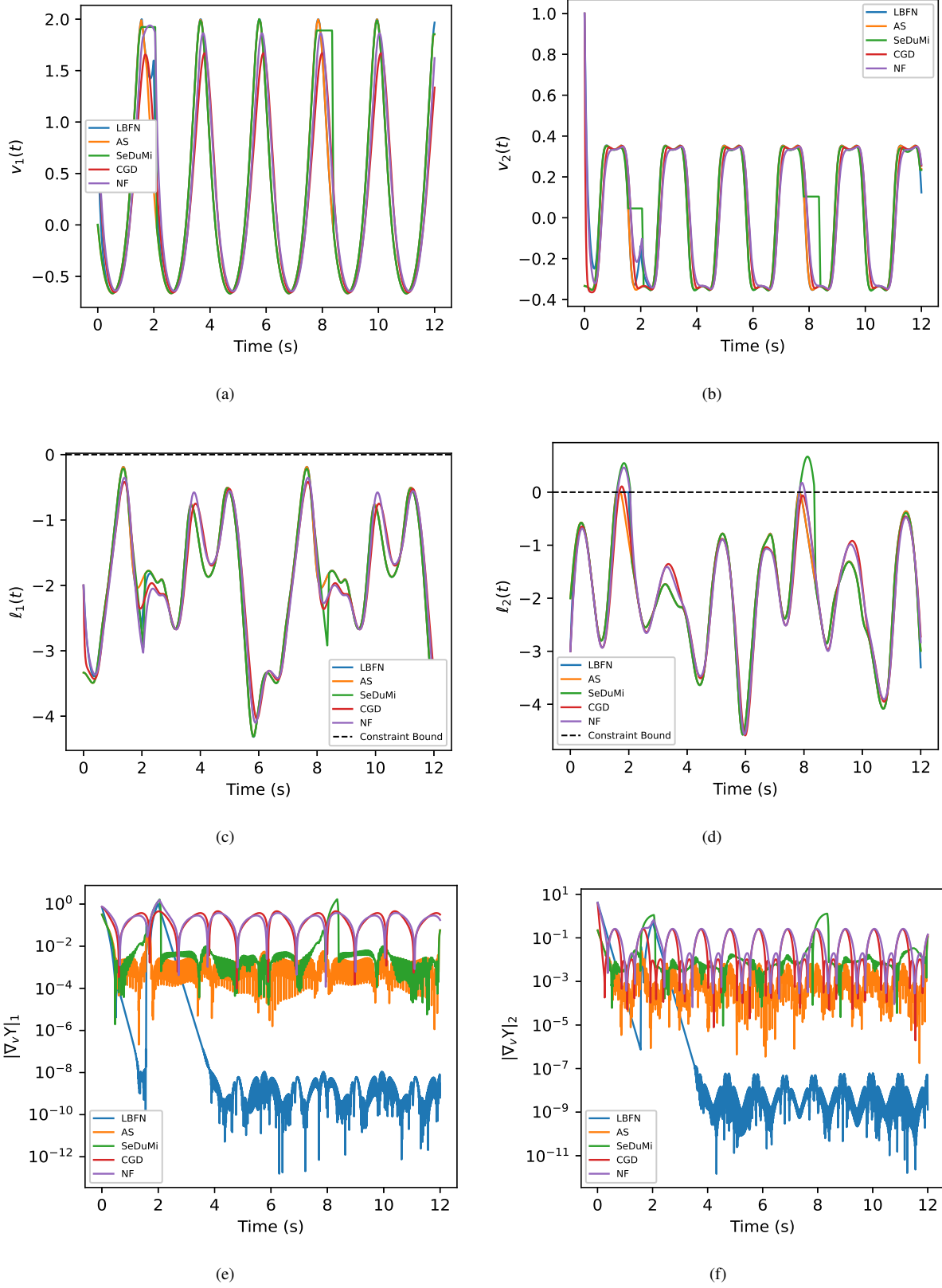
which instantiates the generic problem (1) with  $\mathcal{M}(t) \equiv \mathcal{P}(t)$  and  $\mathbf{g}(t) \equiv \mathbf{n}(t)$ . Notice that the Hessian Matrix  $\mathcal{P}(t)$  has a diagonal structure ensuring the convexity. To be specific,  $\mathcal{P}_{11}(t) = 1 + 0.5 \sin(3t)$  indicates the periodic variation between  $[0.5, 1.5]$ , and  $\mathcal{P}_{22}(t) = 2 + \cos(6t)$  indicates the higher frequency variation between  $[1, 3]$ . Moreover, the eigenvalues of Hessian Matrix  $\mathcal{P}(t)$  are  $\lambda_1(t) \in [0.5, 1.5]$  and  $\lambda_2(t) \in [1, 3]$ . The problem remains strictly convex  $\forall t \in \mathbb{R}$  because:

$$\begin{aligned} \mathcal{P}(t) &\succ 0 \\ \det(\mathcal{P}(t)) &= (1 + 0.5 \sin(3t))(2 + \cos(6t)) \geq 0.5 > 0 \end{aligned}$$



**Figure 2.** Simulative outcomes of the presented LBFN model (9) to solve the optimal solution trajectory of the DQP-IC problem (22), under 10 different starting points (SP9 with  $\mathbf{v}_i(0) \in [-1.5, 1.5]$  sampled at random, plus one deliberately infeasible starting point  $\mathbf{v}(0) = (-3, 4)$ ), with the convergent parameter  $\alpha$  set to 20,  $\gamma_w = 10$  and  $\gamma_r = 5$ . (a) Convergence behaviors of  $v_1(t)$ . (b) Convergence behaviors of  $v_2(t)$ . (c) Constraint behaviors of  $g_1(t) = \sin(2t)v_1(t) + v_2(t) - (2 + \cos(2t))$ . (d) Constraint behaviors of  $g_2(t) = -\cos(2t)v_1(t) + (2t/10)v_2(t) - (2 - \sin(4t))$ . (e) The first entry of the absolute gradient error. (f) The second entry of the absolute gradient error.





**Figure 3.** Comparative simulation among the presented LBFN model (9), AS Approach, Sedumi Approach, CGD Approach and NF Approach for solving the optimal solution trajectory of the DQP-IC problem (22), with  $\alpha = 10$ ,  $\gamma_w = 10$  and  $\gamma_r = 5$ . (a) Convergence behaviors of  $v_1(t)$ . (b) Convergence behaviors of  $v_2(t)$ . (c) Constraint behaviors of  $g_1(t) = \sin(2t)v_1(t) + v_2(t) - (2 + \cos(2t))$ . (d) Constraint behaviors of  $g_2(t) = -\cos(2t)v_1(t) + (2t/10)v_2(t) - (2 - \sin(4t))$ . (e) The first entry of the absolute gradient error. (f) The second entry of the absolute gradient error.



This ensures the strictly convex property of the objective function. Meanwhile, this also ensures a unique global minimum exists at each time instant.

Furthermore, the dynamic inequality constraints  $\mathcal{G}(t)\mathbf{v}(t) \leq \mathbf{h}(t)$  are defined by:

$$\begin{bmatrix} \sin(2t) & 1 \\ -\cos(2t) & 2t/10 \end{bmatrix} \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} \leq \begin{bmatrix} 2 + \cos(2t) \\ 2 - \sin(4t) \end{bmatrix} \quad (21)$$

In this case, this specific example is shown as

$$\begin{aligned} \underset{\mathbf{v}(t) \in \mathbb{R}^n}{\text{minimize}} \quad & \tilde{h}(\mathbf{v}(t), t) = \frac{1}{2} \mathbf{v}^\top(t) \mathcal{P}(t) \mathbf{v}(t) + \mathbf{n}^\top(t) \mathbf{v}(t) \\ \text{subject to} \quad & \ell(\mathbf{v}(t), t) : \mathcal{G}(t) \mathbf{v}(t) \leq \mathbf{h}(t) \end{aligned} \quad (22)$$

Convergence characteristics are comprehensively evaluated under varying convergence parameter  $\alpha$ , as shown in Figure 1, which reveals critical dynamics governing the system behavior. At  $\alpha = 2$ , both variables  $v_1(t)$  and  $v_2(t)$  exhibit persistent oscillations that have not settled even by  $t = 3$  s, where the gradient-error norm remains of order 1 ( $9.2 \times 10^{-1}$ ), only decaying to  $1.7 \times 10^{-2}$  by  $t = 5$  s. Increasing  $\alpha$  sharply accelerates settling: at  $t = 0.2$  s the gradient-error norm is  $8.4 \times 10^{-2}$  for  $\alpha = 20$  but only  $5.1 \times 10^{-5}$  for  $\alpha = 5000$ , and by  $t = 1$  s every case with  $\alpha \geq 20$  has already fallen below  $2.3 \times 10^{-7}$ , confirming the theoretically predicted exponential convergence rate proportional to  $\alpha$ . By  $t = 5$  s, all cases with  $\alpha \geq 20$  converge to a common floor between  $3 \times 10^{-10}$  and  $2 \times 10^{-9}$ , indicating that the long-run floor visible in Figures 1(e)–(f) is governed by the fixed numerical precision of the barrier evaluation rather than by  $\alpha$  itself once  $\alpha$  is large enough for convergence to complete well within the simulated window. All five trajectories additionally show a brief, shared transient rise in gradient error around  $t \approx 1.6$ – $2.0$  s, coinciding with a moment when the time-varying constraint boundary sweeps close to the tracked trajectory (visible as small excursions of  $\ell_1(t)$  and  $\ell_2(t)$  toward zero in Figures 1(c)–(d) at the same time), before the exponential decay resumes. In practice,  $\alpha$  trades off settling speed against numerical stiffness: excessively large  $\alpha$  tightens the ODE's time constant and may require a correspondingly smaller integration step, a consideration we revisit in Section 5.

Initialization robustness is rigorously tested across ten starting points, as shown in Figure 2: nine sampled at random with  $\mathbf{v}_i(0) \in [-1.5, 1.5]$ , plus one deliberately infeasible starting point  $\mathbf{v}(0) = (-3, 4)$  for which  $\mathbf{g}(0) = (1, 1) > \mathbf{0}$  violates both constraints at  $t = 0$  by a margin of 1.0. The LBFN model restores feasibility for this infeasible case within 0.05 s ( $\mathbf{g}(0.05) = (-1.76, -0.37)$ ), and the worst-case gradient-error norm across all ten starting points falls to  $4.5 \times 10^{-2}$  by  $t = 0.3$  s and to  $4.7 \times 10^{-7}$  by  $t = 1$  s. As in Figure 1, a brief transient rise occurs around  $t \approx 2$  s as the moving constraint boundary sweeps past several trajectories simultaneously (worst-case gradient-error norm 1.3, worst-case constraint value  $2 \times 10^{-4}$ , i.e., an essentially negligible boundary touch), after which convergence resumes and the worst-case gradient-error norm falls to  $3.1 \times 10^{-9}$  by  $t = 3$  s – a reduction of 8.9 to 9.7 orders of magnitude from each trajectory's initial value. This establishes the approach's robustness against both initialization variance and initial infeasibility, consistent with Lemma 1 and Corollary 1.

Comparative analysis against established solvers revealed the LBFN model's superior performance in dynamic programming problems, as shown in Figure 3. The comparative approaches include the active-set (AS) approach<sup>19</sup>, the continuous gradient descent (CGD) approach in<sup>35</sup>, the SQP approach in<sup>20</sup>, the SeDuMi approach in<sup>36</sup>, and the Newton flows (NF) approach<sup>37</sup>. To be specific, the SeDuMi and NF approaches show the largest constraint violations, peaking at  $6.7 \times 10^{-1}$  and  $4.7 \times 10^{-1}$  respectively, during two transition windows ( $t \in [1.6, 2.1]$  s and  $t \in [7.8, 8.4]$  s); the AS and CGD approaches show smaller, single-window violations (peaking at  $3.7 \times 10^{-3}$  and  $1.1 \times 10^{-1}$  respectively, near  $t \approx 1.6$ – $1.8$  s), and the CGD approach additionally develops a persistent steady-state offset rather than decaying (median gradient-error norm  $2.5 \times 10^{-1}$  over  $t \in [2, 12]$  s); the LBFN model's own trajectory only briefly touches the constraint boundary during the same transition windows, with a peak excursion of  $5.3 \times 10^{-3}$  – two orders of magnitude smaller than SeDuMi/NF and consistent with the numerical precision of the barrier evaluation (Section 3.1) rather than a genuine, sustained loss of feasibility. Gradient-error convergence metrics confirm distinct advantages: by  $t = 12$  s the LBFN model's gradient-error norm has decayed to  $1.4 \times 10^{-9}$  (median  $6.7 \times 10^{-9}$  over  $t \in [2, 12]$  s), while the AS, SeDuMi, CGD, and NF baselines remain at  $1.3 \times 10^{-1}$ ,  $1.3 \times 10^{-1}$ ,  $3.5 \times 10^{-1}$ , and  $2.2 \times 10^{-1}$  respectively at the same instant – four to five orders of magnitude higher. These results confirm the theoretical predictions of global convergence with tunable rates and persistent feasibility.

## 4.2 Application on Dynamic Robot Navigation

In this part, to further verify the effectiveness of the constructed LBFN model, a dynamic robot navigation task is designed, in which the robot is required to track the dynamic target in real time. The task is formulated as an inequality-constrained dynamic program with a non-quadratic (Gaussian repulsive potential) objective, to which the general LBFN dynamics of Section 2.2

apply directly:

$$\begin{aligned} & \underset{\mathbf{p}(t) \in \mathbb{R}^2}{\text{minimize}} \quad \frac{1}{2} k_t \|\mathbf{p}(t) - \mathbf{p}_{Tar}(t)\|^2 + \sum_i \exp \left( -\frac{\|\mathbf{p}(t) - \mathbf{p}_{Obs_i}(t)\|^2}{k_i} \right), \\ & \text{subject to} \quad \mathcal{A} \mathbf{p}(t) \leq \mathbf{d} \end{aligned} \quad (23)$$

where  $\mathbf{p}(t) \in \mathbb{R}^2$  denotes the robot's planar position (a specific instance of the decision variable  $\mathbf{v}(t)$  in (1) for  $n = 2$ ),  $\mathcal{A} = [1 \ 0; 0 \ 1; -1 \ 0; 0 \ -1]$  and  $\mathbf{d} = [100; 100; 0; 0]$ , and this inequality denotes the physical bounds of the map of the robot navigation task. In this experiment, the constructed LBFN model  $\alpha$  convergent parameter is set to 1000, guaranteeing the swiftly convergent performance and meeting the real-time control for the dynamic robot navigation task (23). Moreover, the moving target's trajectory is set  $\mathbf{p}_{Tar}(t) = [50 + 40 \sin(0.25t), 50 + 25 \cos(0.25t)]^\top$ . The 7 obstacles  $\mathbf{p}_{Obs_i}(t)$  with  $i = 1, 2, \dots, 7$ , separately located at [20; 25] m, [60; 20] m, [70; 60] m, [50; 85] m, [30; 72] m, [38; 35] m and [90; 40] m. The parameters for the obstacles are set  $k_1 = 0.8$ ,  $k_2 = 0.6$ ,  $k_3 = 1.0$ ,  $k_4 = 1.2$ ,  $k_5 = 1.2$ ,  $k_6 = 0.8$  and  $k_7 = 0.9$ , and the parameter for the target  $k_t$  is set as 0.3. The whole task time duration is set as  $T = 30$  s. As mentioned above, the parameter settings are provided.

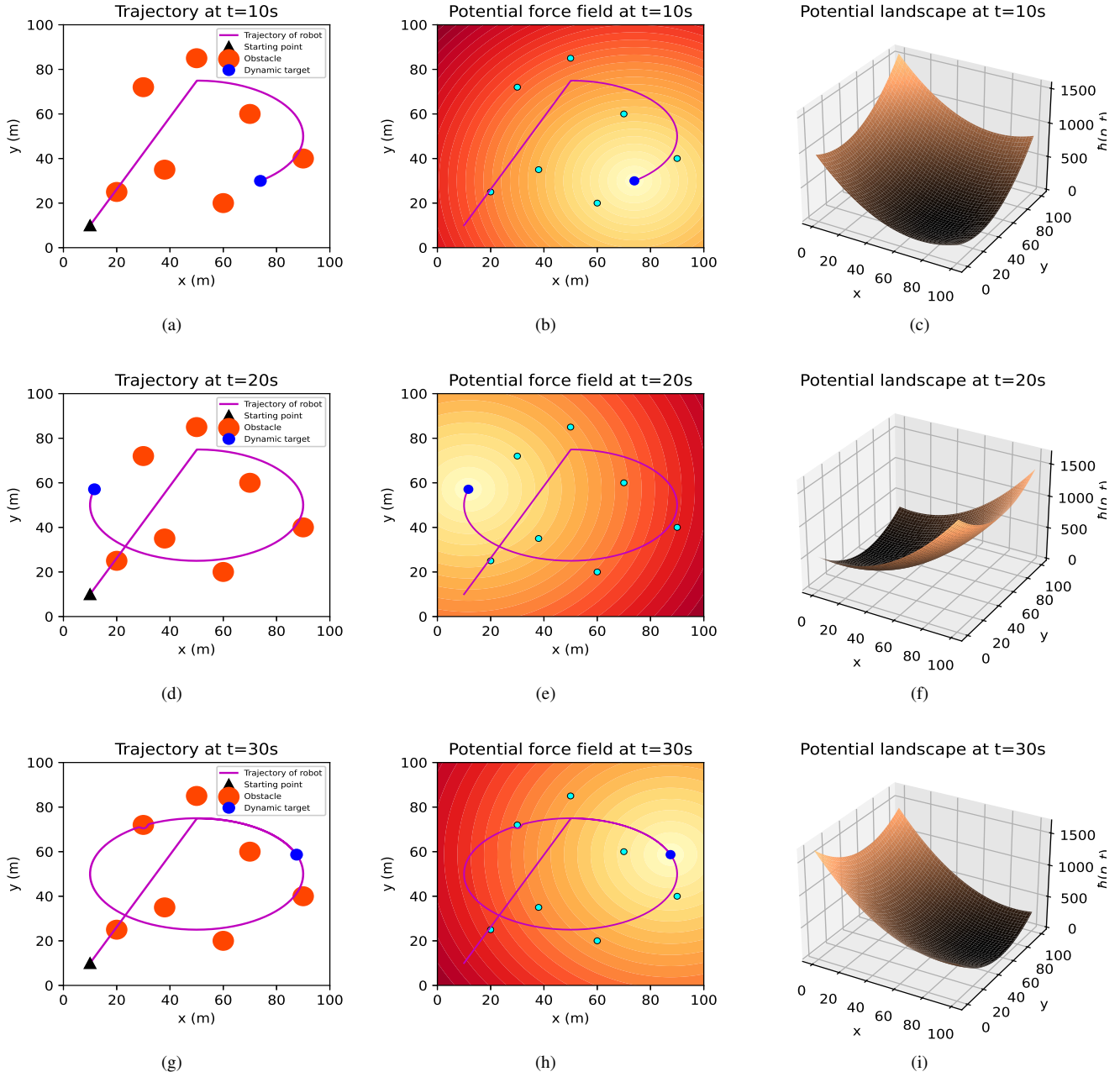
As depicted in Figure 4, the snapshots at the time  $t = 10$  s,  $t = 20$  s, and  $t = 30$  s are offered to present the experimental results. The dynamic target is observed to move along an elliptical pattern trajectory, with seven stationary obstacles involved. Figures 4(a), (d), and (g) provide the intuitive trajectory tracking result, where the robot starts from the starting point (initially 76.3 m from the moving target) and swiftly converges to the moving target, closing to within 0.01 m by  $t = 1$  s and effectively 0 m by  $t = 5$  s, and then follows the moving target in real time, while avoiding the obstacles located in the map (the closest robot-obstacle approach over the entire 30 s run is 1.53 m, near obstacle 5 at  $t \approx 23$  s, and the robot remains within the map's  $[0, 100] \times [0, 100]$  bounds throughout). Further, Figures 4(b), (e), and (h) provide the potential force field on contour, and Figures 4(c), (f), and (i) provide the potential function landscape, for different time instants. This intuitively demonstrates that under the control of the constructed LBFN model, the robot can excellently accomplish the given dynamic robot navigation task in real time.

## 5 Limitations and Future Work

While the LBFN model demonstrates strong theoretical guarantees and empirical performance, several limitations delineate directions for future research. First, although Section 2.2 shows that Lemma 1, Theorem 1, and Corollary 1 extend verbatim from the quadratic case (1) to any objective that is  $m_f$ -strongly convex in  $\mathbf{v}(t)$  for every  $t$ , this still excludes objectives that are only *locally* convex, such as the Gaussian repulsive potential used in Section 4.2, whose Hessian can lose positive definiteness near an obstacle; extending the convergence guarantees to such locally nonconvex, or more generally nonconvex, dynamic programs would broaden applicability but requires new arguments beyond the strong-convexity-based Lyapunov analysis of Section 3. Second, the model assumes that  $\mathcal{M}(t)$ ,  $\mathbf{g}(t)$ ,  $\mathcal{N}(t)$ ,  $\mathbf{d}(t)$  and their time derivatives are known in closed form; in practice these quantities are often estimated from noisy sensor data, so coupling the LBFN dynamics with derivative estimators or observer-based prediction would be necessary for deployment on physical hardware. Third, although Section 3.1 shows the per-instant cost scales as  $O(n^3 + pn^2)$ , this remains cubic in the state dimension  $n$ ; for large-scale problems, exploiting the Sherman–Morrison–Woodbury identity for the rank- $p$  Hessian update, warm-started or incremental Cholesky factorizations across nearby time instants, or distributed/parallel implementations across constraint blocks are promising ways to reduce this cost further. Fourth, the barrier parameter  $w(t) = w(0)e^{\gamma_w t}$  grows without bound, which is necessary for asymptotic exactness but can induce numerical ill-conditioning of  $\nabla_{\mathbf{v}\mathbf{v}} Y$  near the constraint boundary over long horizons; adaptive or saturating barrier schedules merit investigation for long-duration deployments. Finally, the robot navigation case study in Section 4.2 is validated in simulation; hardware-in-the-loop and physical robot experiments, together with benchmarking on higher-dimensional and multi-agent DQP-IC instances, are natural next steps to further substantiate the real-time applicability of the proposed model.

## 6 Conclusions

This paper has established a logarithmic barrier function-based Newton's (LBFN) model for real-time solving of dynamic quadratic programming problems subject to inequality constraints. The proposed model fundamentally advances beyond traditional approaches by employing parametric adaptive log-barrier functions to directly handle inequality constraints, thereby avoiding the need for auxiliary variables typically required in KKT-based methods. We established the well-posedness of the resulting continuous-time dynamics, proved global asymptotic stability with an exponentially decaying, tunable tracking error via a Lyapunov argument, and derived a per-instant computational complexity of  $O(n^3 + pn^2)$  that is comparable to a single Newton step on a static quadratic program of the same size. These properties make the model particularly suitable for dynamic environments where both computational efficiency and constraint adherence are critical. Numerical experiments comprehensively validated the model's performance, showing superior convergence speed and solution accuracy compared to



**Figure 4.** Experimental results of the dynamic robot navigation task (23) accomplished by the constructed LBFN model in real time, with the task duration being  $T = 30$  s, involving 7 obstacles. (a) Robot motion trajectory at  $t = 10$  s. (b) Potential force field on contour at  $t = 10$  s. (c) Potential function landscape at  $t = 10$  s. (d) Robot motion trajectory at  $t = 20$  s. (e) Potential force field on contour at  $t = 20$  s. (f) Potential function landscape at  $t = 20$  s. (g) Robot motion trajectory at  $t = 30$  s. (h) Potential force field on contour at  $t = 30$  s. (i) Potential function landscape at  $t = 30$  s.

active-set, sequential quadratic programming, interior-point, continuous gradient-descent, and Newton-flow baselines across various operational scenarios, while a dynamic robot navigation case study further demonstrated simultaneous trajectory tracking and obstacle avoidance in real-time operation. Building on the limitations discussed in Section 5, future work will extend the LBFN framework to nonlinear and nonconvex dynamic programs, incorporate noise-robust parameter estimation, exploit factorization-update techniques for large-scale problems, and pursue hardware validation on physical robotic platforms.

## Data availability

The Python implementation of the LBFN model and all baseline solvers, together with the scripts and data that reproduce the numerical experiments and figures reported in this study, are available in the GitHub repository at <https://github.com/yzhou364/LBFN>.

## Author contributions

Y.Z. conceived the research idea, supervised the study, and revised the manuscript; Y.Z. is the corresponding author. J.W. developed the theoretical framework, implemented the LBFN algorithm, conducted the numerical simulations and the robot navigation experiment, and drafted the manuscript. Both authors reviewed and approved the final version of the manuscript.

## Use of AI

An AI assistant (Claude, Anthropic) was used to assist with implementing the numerical simulation code, carrying out the numerical experiments reported in Section 4, and drafting and revising the manuscript text. The authors designed the study, verified the correctness of the implementation and results, and take full responsibility for the content of this manuscript.

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## Competing interests

The authors declare no competing interests.

## Additional information

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